A CASE-STUDY BASED REPORT

ON

Customer Segmentation Using K-Means and Hierarchical Clustering

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by

AMGOTH BHAVIKA PAWAR	**2520030616**
DUMPALA HEMASAI	**2520030599**
MUTHANA NIKHIL REDDY	**2520030560**

Under the Esteemed guidance of

Ravindra gaddam

Assistant Professor

Dept. of CSE

Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

Koneru Lakshmaiah Education Foundation
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ABSTRACT:

Customer segmentation is an important technique in data analytics that enables businesses to understand customer characteristics, purchasing patterns, and spending behavior. This project, titled Customer Segmentation Using K-Means and Hierarchical Clustering, aims to group customers into meaningful segments based on their annual income and spending score using unsupervised machine learning techniques.The project uses a customer dataset and performs data preprocessing, exploratory data analysis, feature scaling, and clustering. The K-Means algorithm divides customers into a predefined number of clusters by assigning them to the nearest cluster centroid. Hierarchical Clustering groups similar customers through a sequence of cluster-merging operations, which can be visualized using a dendrogram. The Elbow Method is used to help determine a suitable number of clusters, while the Silhouette Score is used to evaluate cluster separation and cohesion.

The project also includes visualizations for understanding customer groups and an interactive interface developed using Gradio. This interface allows users to enter annual income and spending score values to obtain a predicted customer segment and a corresponding marketing suggestion. The results can help businesses identify different customer categories, design targeted marketing strategies, improve customer engagement, and support data-driven decision-making. Overall, this project demonstrates the practical application of unsupervised machine learning in customer analytics. The collected data is first preprocessed by handling missing values, scaling numerical features, and selecting relevant attributes. K-Means clustering partitions customers into a fixed number of clusters by minimizing the distance between data points and cluster centroids, while Hierarchical Clustering builds a tree-like structure (dendrogram) to reveal the relationships among customers at different levels of similarity. The performance and effectiveness of both algorithms are analyzed and compared using visualization techniques and clustering evaluation metrics. Customer segmentation is an important technique that helps businesses understand the diverse characteristics and purchasing behaviors of their customers.

This project, **"Customer Segmentation Using K-Means and Hierarchical Clustering,"** aims to group customers into meaningful segments based on their demographic and behavioral data, such as age, annual income, and spending score. The project utilizes two popular unsupervised machine learning algorithms—**K-Means Clustering** and **Hierarchical Clustering**—to identify natural patterns within the customer dataset without requiring predefined class labels. The outcome of this project enables businesses to identify customer groups with similar characteristics, allowing them to design targeted marketing strategies, improve customer satisfaction, optimize product recommendations, and increase profitability. Overall, this project demonstrates how unsupervised machine learning techniques can transform raw customer data into valuable business insights, supporting better decision-making and personalized customer engagement.

1. INTRODUCTION:

Customer behavior plays a significant role in business growth and marketing strategy. Every business serves customers with different income levels, purchasing habits, preferences, and spending patterns. Treating all customers in the same way may result in ineffective marketing campaigns and missed business opportunities. Therefore, identifying groups of customers with similar characteristics is an important task.

Customer segmentation is the process of dividing a customer population into smaller groups based on shared characteristics. Machine learning provides effective techniques for performing this task automatically. In particular, unsupervised learning algorithms can discover patterns in customer data without requiring predefined category labels.

This project applies K-Means and Hierarchical Clustering to customer data to identify meaningful customer groups. The primary features used for clustering are annual income and spending score. The project also uses data visualization and clustering evaluation techniques to examine the resulting groups and understand their characteristics.

1.1 Background:

The increasing availability of customer data has created opportunities for businesses to understand their customers more effectively. Traditional approaches often rely on broad assumptions or manually defined customer categories. These approaches may not capture complex relationships within large datasets.

Machine learning techniques can analyze numerical customer attributes and identify groups based on similarity. K-Means is a widely used partition-based clustering algorithm that assigns observations to clusters around their centroids. Hierarchical Clustering constructs a hierarchy of clusters by repeatedly merging similar observations or groups.

By applying these algorithms to customer income and spending information, businesses can identify patterns such as high-income customers with high spending scores, customers with lower spending scores, and customers with moderate spending behavior. These insights can support customer relationship management and marketing decisions.

1.2 Motivation for the Project:

The main motivation for this project is to demonstrate how machine learning can convert raw customer data into useful business insights. Businesses need to understand which customer groups are more likely to respond to particular offers, products, and promotional campaigns.

Manual segmentation can be time-consuming and may not reveal patterns that exist across multiple customer attributes. An automated clustering system provides a systematic way to explore these patterns.

This project also provides an opportunity to study and compare two different unsupervised learning algorithms. By examining their cluster assignments, visualizations, and evaluation scores, the project demonstrates how different clustering approaches can be used for the same business problem.

1.3 Overview of the Work Carried Out:

The project follows a structured machine learning workflow. First, the customer dataset is loaded into a Python environment and examined to understand its structure, attributes, and quality. Next, data preprocessing is performed, and the relevant numerical features are selected.

Exploratory Data Analysis is then conducted using statistical summaries and visualizations to understand the distributions and relationships among customer attributes. The selected features are standardized to ensure that their numerical scales do not disproportionately influence distance calculations.

The Elbow Method is used to investigate a suitable number of clusters. K-Means and Hierarchical Clustering are subsequently applied to identify customer groups. The resulting clusters are examined using scatter plots, centroid summaries, cluster-size analysis, and the Silhouette Score.

Finally, an interactive Gradio interface is provided so users can enter customer attributes and obtain a predicted cluster and a suggested marketing strategy.

1.4 Scope and Responsible-Use Statement:

The scope of this project is limited to exploring customer segmentation using unsupervised machine learning techniques. The main clustering features are annual income and spending score. The project focuses on identifying patterns in the available dataset rather than predicting future purchases or establishing causal relationships.

The system can assist businesses in understanding customer groups and planning targeted marketing activities. However, the identified clusters should be treated as analytical categories, not definitive judgments about individual customers.

The quality of the results depends on the dataset, selected features, preprocessing methods, and clustering parameters. Customer information should be handled responsibly, and the segmentation results should not be used to discriminate unfairly against customers. Business decisions should consider additional context and human judgment.

2. PROBLEM STATEMENT:

Customer segmentation is a challenging task for businesses because customers have different financial characteristics and spending behaviors. When customers are treated as a single group, businesses may struggle to identify their needs, recognize high-value customer segments, and develop effective marketing strategies.

Traditional methods of customer classification may depend on predefined rules and may not identify natural groupings within customer data. Therefore, a data-driven approach is required to analyze customer attributes and discover meaningful patterns automatically.

This project addresses the problem by applying K-Means and Hierarchical Clustering to customer data containing annual income and spending score. The project aims to identify distinct customer groups, visualize their characteristics, and evaluate the quality of the resulting clusters.

2.1 Case-Study Scenario:

Consider a retail business that maintains information about its customers, including annual income and spending score. The business wants to understand how customers differ in their spending behavior so that it can improve its marketing strategies.

Some customers may have high annual incomes and high spending scores, while others may have high incomes but lower spending scores. Similarly, some customers may have lower incomes but relatively high spending scores.

Without a systematic analysis, identifying these patterns across a large customer population can be difficult. The business therefore requires a method that can automatically group similar customers and present the results in an understandable format.

In this project, K-Means and Hierarchical Clustering are applied to the customer dataset. Their outputs are analyzed through visualizations and evaluation metrics to understand the identified customer groups. The resulting insights can support the development of customer-specific promotional campaigns and marketing strategies.

2.2 Central Case-Study Question:

The central question addressed by this project is:

How can unsupervised machine learning algorithms, particularly K-Means and Hierarchical Clustering, be used to identify meaningful customer segments based on annual income and spending score, and how can these segments support better business and marketing decisions?

The project also investigates the following supporting questions:

- What patterns and relationships exist in the available customer data?
- How can the appropriate number of customer clusters be estimated?
- How do K-Means and Hierarchical Clustering identify customer groups?
- How can clustering quality be evaluated using the Silhouette Score?
- How can the resulting customer segments be presented through visualizations and an interactive interface?

2.3 Formal Problem Definition:

Let the customer dataset be represented by a collection of observations:

$$ D=\{x_1,x_2,\ldots,x_n\} $$

Each observation represents a customer and is described using two primary numerical features: annual income and spending score.

The objective is to partition the observations into K clusters such that customers within the same cluster have similar feature values, while customers belonging to different clusters are comparatively dissimilar.

For K-Means, the objective is to minimize the within-cluster sum of squared distances:

$$ J=\sum_{k=1}^{K}\sum_{x_i\in C_k} \|x_i-\mu_k\|^2 $$

where:

- K is the number of clusters.
- C_k is the set of customers assigned to cluster k.
- μ_k is the centroid of cluster k.
- x_i is the feature vector representing a customer.

Hierarchical Clustering addresses the same segmentation problem using a hierarchical merging process. In this project, agglomerative clustering with Ward linkage is used to form five clusters.

The overall objective is to identify interpretable customer segments, compare the clustering approaches, and present the results in a form that can support business analysis.

2.4 Objectives that Guide the Study:

The major objectives of this project are:

1. To study customer characteristics using a structured customer dataset.
2. To perform data preprocessing and exploratory data analysis.
3. To select annual income and spending score as the primary clustering features.
4. To standardize the selected features before applying distance-based algorithms.

5. To estimate a suitable number of clusters using the Elbow Method.
6. To implement K-Means and Hierarchical Clustering for customer segmentation.
7. To visualize the resulting clusters and analyze their characteristics.
8. To evaluate clustering quality using the Silhouette Score.
9. To compare the behavior and practical advantages of the two algorithms.
10. To develop an interactive interface for predicting a customer's cluster and displaying a suggested marketing strategy.

2.5 Assumptions and Limitations:

Assumptions:

- The dataset contains customer attributes that are sufficiently representative for the intended analysis.
- Annual income and spending score provide useful information for distinguishing customer groups.
- Customers with similar values for these features may exhibit similar spending-related characteristics.
- Standardization makes the selected features more comparable for distance-based clustering.
- Five clusters provide a useful segmentation for the project's analysis, based on the Elbow Method and the chosen implementation.

Limitations:

- The quality of the segmentation depends on the accuracy, completeness, and representativeness of the dataset.
- The model primarily uses annual income and spending score, so it does not capture every factor affecting customer behavior.
- K-Means can be sensitive to initialization and the selected number of clusters.
- Hierarchical Clustering may require substantial computational resources for large datasets; the project therefore uses a sample for this analysis.
- Cluster labels are generated by the algorithms and do not automatically represent predefined business categories.
- The Silhouette Score measures cluster separation and cohesion but does not establish whether the resulting groups will increase business profitability.
- The interactive interface predicts a cluster using the trained model; it does not guarantee a customer's future purchases or spending behavior.

3. DATASET AND EXPLORATORY DATA ANALYSIS:

3.1 About the Dataset:

The project uses a customer dataset stored in CSV format. The data is loaded into Python using the Pandas library and examined to understand its structure and suitability for machine learning.

The dataset includes customer-related attributes used for analyzing spending behavior. The two primary features selected for clustering are annual income and spending score.

Feature	Description	Role in the project
Annual Income (k$)	Annual income represented in thousands of dollars	Primary clustering feature
Spending Score (1–100)	Numerical score representing customer spending behavior	Primary clustering feature
Age	Customer's age, where available in the dataset	Additional analysis
Purchase frequency	Measure of how frequently a customer purchases, if available	Additional analysis
Customer ID	Identifier used to distinguish records	Identification, not a primary clustering feature

The exact availability of additional attributes depends on the CSV file. The main clustering analysis is based on annual income and spending score.

The dataset is suitable for demonstrating unsupervised learning because the algorithms can discover customer groups without requiring a predefined target column containing customer-segment labels.

3.2 Purpose of EDA:

Exploratory Data Analysis (EDA) is performed to understand the characteristics, distribution, and relationships of the available customer data before applying machine learning algorithms.

The main purposes of EDA in this project are:

- To examine the dataset's dimensions, column names, and data types.
- To identify missing values and duplicate records.
- To understand the distributions of annual income and spending score.
- To examine relationships between customer attributes.
- To identify potential patterns and unusual observations.
- To support the selection of suitable features for clustering.
- To visualize customer data before and after segmentation.

EDA helps ensure that the clustering process is based on a clear understanding of the data rather than applying algorithms without first examining the dataset.

3.3 Preparation Steps Actually Performed:

The dataset is prepared before applying the clustering algorithms. The following steps describe the preprocessing workflow used in the project.

1. Loading the dataset:

The CSV file is loaded using Pandas. The dataset's dimensions, column names, data types, and sample records are examined.

```
dataset = pd.read_csv(file_name)

print(dataset.head())
print(dataset.shape)
print(dataset.info())
```

2. Checking data quality:

The dataset is examined for missing values and duplicate records. This helps identify potential data-quality issues that could affect the analysis.

```
print(dataset.isnull().sum())
print(dataset.duplicated().sum())
```

3. Selecting relevant features:

Annual income and spending score are selected as the primary features for clustering.

```
base_features = [
    'Annual Income (k$)',
    'Spending Score (1-100)'
]

X_raw = dataset[base_features].values
```

4. Standardizing the features:

StandardScaler is applied to transform the selected features to a comparable scale.

```
scaler = StandardScaler()
X = scaler.fit_transform(X_raw)
```

Standardization is important because the clustering algorithms rely on distance calculations. It prevents differences in numerical scale from disproportionately influencing the results.

5. Preparing data for clustering:

The standardized data is supplied to the clustering algorithms. The project uses the complete dataset for K-Means and a sample of up to 5,000 observations for Hierarchical Clustering.

These steps prepare the customer data for clustering and subsequent evaluation.

3.4 EDA Exploration – Implementation:

The EDA stage uses Python libraries such as Pandas, Matplotlib, and Seaborn to explore the dataset and display its characteristics.

Statistical analysis:

Descriptive statistics are used to examine the central tendency and variation of numerical attributes.

```
print(dataset.describe())
```

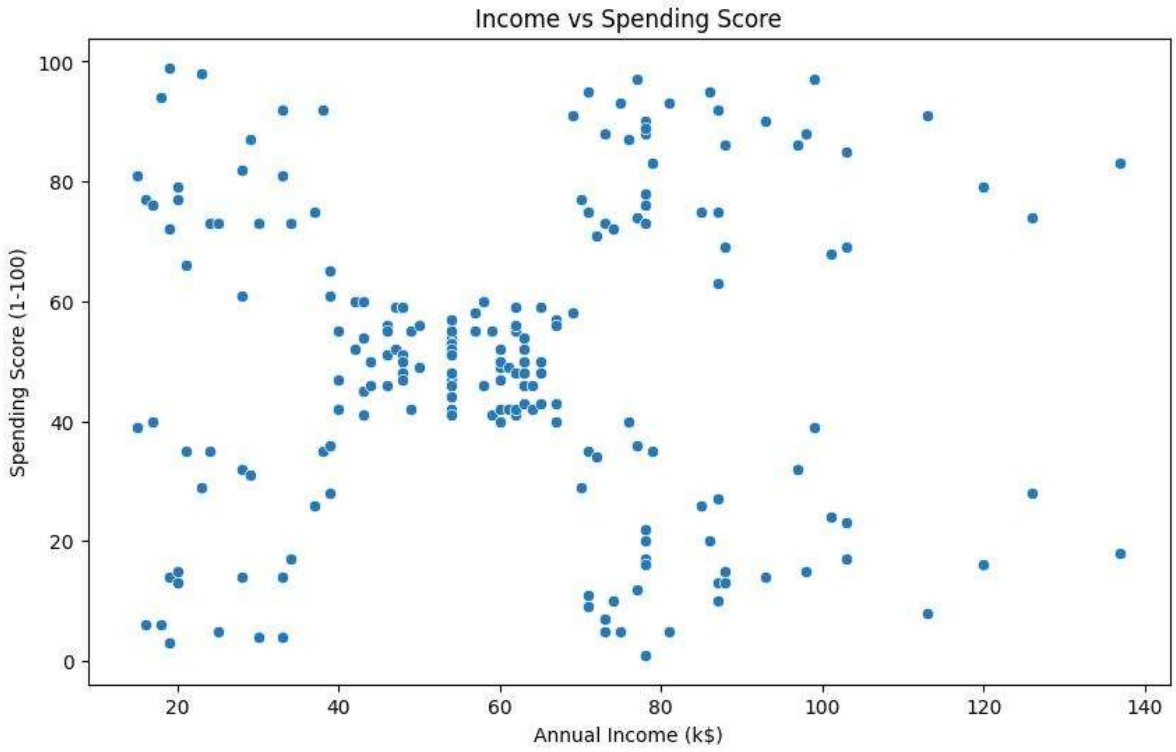
This helps identify typical income and spending values, the range of the observations, and potential variations among customers.

Distribution analysis:

Histograms and related visualizations can be used to examine the distribution of annual income and spending scores. These plots help identify concentration patterns, variation, and possible unusual observations.

Relationship analysis:

A scatter plot of annual income against spending score helps reveal how customers are distributed across the two-dimensional feature space.



Clustering visualizations:

After training the models, scatter plots display the resulting customer groups. For K-Means, the cluster centroids are also plotted to show the central positions of the groups.

For Hierarchical Clustering, a dendrogram illustrates the sequence of cluster merges. Additional plots can show the number of customers in each cluster and compare the average characteristics of the identified groups.

Evaluation:

The Elbow Method is used to examine the within-cluster sum of squares for different values of \(K\). The Silhouette Score is then used to assess the separation and cohesion of the clusters.

Together, these analyses help interpret the dataset and understand the segmentation results.

3.5 Summary of EDA Findings:

The exploratory analysis provides the foundation for applying machine learning to customer segmentation. It helps establish the structure and characteristics of the customer data and supports the selection of annual income and spending score as the main clustering features.

The scatter plots provide a visual understanding of how customers are distributed across income and spending levels. The analysis also supports the identification of customer groups with different spending-related characteristics.

The Elbow Method helps select a practical cluster count for further analysis. The resulting clustering visualizations and evaluation scores can then be used to examine whether the identified groups are reasonably distinct.

The final interpretation of the EDA should be based on the actual outputs generated by the notebook, including its descriptive statistics, plots, cluster summaries, and evaluation scores.

4. ALGORITHM SELECTION AND IMPLEMENTATION:

4.1 Overview of Algorithm Selection:

The project uses two unsupervised machine learning algorithms: K-Means Clustering and Agglomerative Hierarchical Clustering. Both algorithms can identify groups of similar customers without requiring predefined customer-segment labels.

K-Means is selected because it is computationally efficient and suitable for grouping large numerical datasets. Hierarchical Clustering is selected because it reveals relationships between observations through a hierarchy of cluster merges.

Using both algorithms provides an opportunity to examine customer segmentation from two different perspectives.

4.2 K-Means Clustering:

K-Means is a partition-based clustering algorithm that divides the dataset into a specified number of clusters. Each cluster is represented by a centroid, which is the mean position of the observations assigned to that cluster.

In this project, the number of clusters is set to five. The Elbow Method is used to help select this value.

The implementation follows these steps:

1. Initialize five centroids using the K-Means++ method.
2. Assign each customer to the nearest centroid.
3. Recalculate the centroids based on the assigned customers.

4. Repeat the assignment and centroid-update steps until convergence.
5. Store the final cluster labels and centroid positions.
6. Visualize the customer groups and analyze their characteristics.

The implementation uses Scikit-learn's KMeans class. The fit_predict() method fits the model and returns a cluster assignment for each observation.

The resulting clusters can be interpreted using their centroid coordinates and customer characteristics. These interpretations may support marketing strategies such as loyalty programs, personalized promotions, and targeted product recommendations.

4.3 Hierarchical Clustering:

Hierarchical Clustering is an unsupervised learning algorithm that builds a hierarchy of clusters. This project uses the agglomerative approach, which begins with individual observations and repeatedly merges similar clusters.

The implementation uses Euclidean distance and Ward linkage. Ward linkage selects merges that minimize the increase in within-cluster variance.

The main steps are:

1. Begin with individual observations as separate clusters.
2. Calculate distances between observations or clusters.
3. Merge the closest clusters according to the selected linkage method.
4. Continue merging until five clusters remain.
5. Examine the resulting cluster assignments and dendrogram.

The project uses Scikit-learn's AgglomerativeClustering class. Because hierarchical clustering can be computationally expensive on large datasets, the implementation uses a sample of up to 5,000 observations.

The dendrogram provides a visual representation of the hierarchical relationships between observations and clusters. This helps explain how the groups are formed at different levels of similarity.

4.4 Elbow Method:

The Elbow Method is used to estimate an appropriate number of clusters for K-Means. It calculates the within-cluster sum of squares for different cluster counts.

As the number of clusters increases, the within-cluster sum of squares generally decreases. The preferred cluster count is often near the point where the reduction begins to slow down.

In this project, the Elbow Method examines cluster counts from one to ten and supports the selection of five clusters for the final analysis.

However, the Elbow Method is a heuristic. The selected value should also be considered alongside the cluster visualizations, Silhouette Score, and the intended business application.

4.5 Silhouette Score:

The Silhouette Score is an internal evaluation metric used to assess clustering quality. It compares the distance of an observation to its own cluster with its distance to the nearest alternative cluster.

The score ranges from -1 to +1. A value close to +1 generally indicates that observations are well matched to their assigned clusters. Values near zero indicate overlapping groups, while negative values can indicate potentially inappropriate assignments.

The project calculates Silhouette Scores to evaluate the clustering results. For a fair comparison, the algorithms should be evaluated on the same observations using consistent feature preprocessing.

A higher Silhouette Score can indicate better cluster separation and cohesion, but it does not necessarily mean that the algorithm produces more useful business segments.

4.6 Comparison of the Algorithms:

Comparison factor	**K-Means**	**Hierarchical Clustering**
Clustering approach	Centroid-based partitioning	Agglomerative cluster merging
Number of clusters	Five	Five
Main visualization	Cluster scatter plot with centroids	Cluster scatter plot and dendrogram
Computational efficiency	Generally efficient for large datasets	Can be expensive for large datasets
Main strength	Fast and straightforward to interpret	Shows hierarchical relationships
Main limitation	Sensitive to initialization and cluster count	Higher computational and memory requirements
Evaluation	Silhouette Score	Silhouette Score

K-Means is useful for efficiently assigning customers to a fixed number of groups. Hierarchical Clustering provides additional insight into the relationships between customer observations.

The final comparison should consider the actual evaluation scores, visual separation, cluster sizes, and interpretability rather than assuming that one algorithm is always superior.

4.7 Interactive Prediction Interface:

The project includes an interactive interface developed using Gradio. It allows users to enter a customer's annual income and spending score and obtain a predicted customer segment using the trained K-Means model.

The interface follows this process:

1. Accept annual income and spending score as user inputs.
2. Validate the entered values.
3. Convert annual income into the units expected by the model.
4. Apply the same feature scaling used during training.
5. Use the trained K-Means model to predict the cluster.
6. Display the predicted segment and a corresponding marketing suggestion.

The interface makes the project easier to demonstrate because users can experiment with different customer profiles without manually executing the clustering code.

The prediction represents the cluster assigned by the trained model. It should not be interpreted as a guaranteed prediction of future purchasing behavior.

5. CONCLUSION:

The project, Customer Segmentation Using K-Means and Hierarchical Clustering, demonstrates the application of unsupervised machine learning to customer data analysis. Its primary objective is to identify meaningful customer groups based on annual income and spending score and examine how these groups can support business decision-making.

The project follows a structured workflow involving dataset loading, data preprocessing, feature standardization, exploratory data analysis, cluster selection, model implementation, and evaluation. K-Means is used to partition the customer dataset into five clusters, while Agglomerative Hierarchical Clustering provides an alternative grouping approach and a dendrogram for visualizing cluster relationships.

The Elbow Method helps estimate a suitable number of clusters, and the Silhouette Score provides a measure of cluster separation and cohesion. Visualizations and cluster summaries help interpret the resulting groups and understand their characteristics. The Gradio interface further demonstrates the practical application of the trained K-Means model by allowing users to obtain customer-segment predictions from new input values.

Overall, the project illustrates how machine learning can help transform customer data into interpretable segments that support targeted marketing, customer engagement, and data-driven planning. Future improvements could include incorporating additional customer attributes, testing alternative clustering algorithms, optimizing the number of clusters, and evaluating the stability of the identified segments on different datasets.